ATM Simulation System

Project Report

By

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Acknowledgement

The project **"ATM Simulation System"** is the Project work carried out by

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We are thankful to our project guide for guiding us to complete the Project.

Their suggestions and valuable information regarding the formation of the Project Report have provided us a lot of help in completing the Project and its related topics.

Abstract

Managing banking transactions efficiently and securely is a crucial requirement in today's digital world. The Smart ATM Simulation System is a Java-based application developed using Swing UI and SQLite/MySQL database to simulate real-world ATM operations.
The system provides both traditional and modern banking features such as:

- Card-based login with PIN authentication
- Mobile number + PIN login
- QR-based UPI withdrawal
- OTP-based cardless withdrawal
- Fingerprint authentication (simulated)
- Multi-account handling (Savings, Current, Salary)

The system also implements security mechanisms such as SHA-256 PIN encryption and OTP verification. The user interface mimics a real ATM machine with left and right button navigation, providing a realistic user experience.
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1. Introduction

Automated Teller Machines (ATMs) are widely used banking systems that allow customers to perform financial transactions without visiting a bank.

Traditional ATM systems are limited to card-based operations and basic functionalities such as withdrawal, deposit, and balance inquiry. With the rise of digital banking, modern ATMs now support advanced features such as UPI withdrawals, biometric authentication, and cardless transactions.

This project aims to develop a **ATM Simulation System** that integrates both traditional and modern banking features into a single platform using Java.

1. Background of the Study

ATMs play an important role in banking by providing 24/7 access to financial services. However, traditional systems have limitations such as:

- Dependence on physical cards
- Limited authentication methods
- Lack of integration with digital payment systems

With advancements in technology, modern ATMs now support QR-based payments and biometric authentication.

2. Motivation

The motivation behind this project includes:

- Understanding real-world ATM system design
- Implementing secure authentication mechanisms
- Integrating modern features like UPI and OTP withdrawal
- Building a realistic ATM user interface

3. Significance of Study

This project is useful for:

- Students learning Java and system design
- Developers understanding banking systems
- Demonstrating real-world application of OOP concepts

4. Features of ATM Simulation System

The ATM Simulation System offers multiple modules that make banking operations efficient, secure, and user-friendly. The system is designed to simulate real-world ATM functionalities along with modern digital banking features.

Authentication Module

a) Allows login using Card + PIN authentication.
b) Supports Mobile Number + PIN login without a card.
c) Provides OTP-based authentication for secure transactions.
d) Supports Fingerprint authentication (simulated) for biometric login.
e) Blocks user access after multiple incorrect PIN attempts.

Account Management

a) Supports multiple account types such as Savings, Current, and Salary accounts.
b) Allows users to select account type before performing transactions.
c) Maintains account balance and transaction history.
d) Ensures accurate and real-time account updates.

Transaction Module

a) Perform cash withdrawal from selected account.
b) Deposit money into account.
c) Check account balance.
d) View mini statement (recent transactions).
e) Change PIN securely.

UPI QR Withdrawal Module

a) Generate QR code for withdrawal instead of entering UPI ID.
b) Allow users to scan QR using mobile UPI apps.
c) Verify payment and process withdrawal.
d) Enable cardless and fast transactions.

OTP Cardless Withdrawal Module

a) Allows withdrawal without ATM card.
b) User enters mobile number linked to account.
c) OTP is generated and verified.
d) Cash is dispensed after successful verification.

Biometric (Fingerprint) Module

a) Simulates fingerprint-based authentication.
b) Stores fingerprint hash securely in database.
c) Allows login without PIN using biometric verification.

Admin Module

a) Admin login with secure credentials.
b) View all users, accounts, and transactions.
c) Add new users and manage existing users.
d) Reset user PIN and block/unblock cards.
e) Monitor ATM cash status and system activity.

ATM Hardware Simulation

a) Card Reader for simulating card insertion.
b) Cash Dispenser for withdrawal operations.
c) Receipt Printer for transaction slips.
d) QR Generator for UPI transactions.
e) Fingerprint Scanner for biometric login.

5. Reporting and Analytics

• Generate transaction history reports for users.
• Display account balance summaries.
• Provide logs of all ATM transactions.
• Allow admin to monitor system usage and activity.

2. System Analysis

System analysis is a key phase in the Software Development Life Cycle (SDLC) that involves identifying system limitations, understanding requirements, and evaluating feasibility.

For the ATM Simulation System, it focuses on improving traditional ATM limitations by introducing secure authentication, cardless transactions, and modern features like UPI QR withdrawal, ensuring a more efficient and user-friendly system.

1. Problem Definition

Traditional ATM systems:

- Require physical cards
- Have limited features
- Do not support modern digital payments

2. Objectives of the System

The main objectives are:

- To simulate a real ATM system
- To implement multiple authentication methods
- To support UPI QR-based withdrawal
- To provide a user-friendly ATM interface
- To ensure secure transactions

3. Feasibility Study

a. Technical Feasibility
- Developed using Java Swing and SQLite/MySQL
- Secure encryption implemented

b. Economic Feasibility
- Uses open-source tools
- No additional cost

c. Operational Feasibility
- Easy to use UI
- Realistic ATM simulation

d. Time Feasibility
- Completed within project timeline

4. System Requirements

Functional Requirements

- Card login
- Mobile login
- PIN authentication
- Cash withdrawal and deposit
- UPI QR withdrawal
- OTP withdrawal
- Fingerprint authentication
- Account selection

Non-Functional Requirements

- Security
- Reliability
- Performance
- Usability

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4GB+ +

Software Requirements

- Java (JDK 17)
- SQLite/MySQL

5. ER diagram:

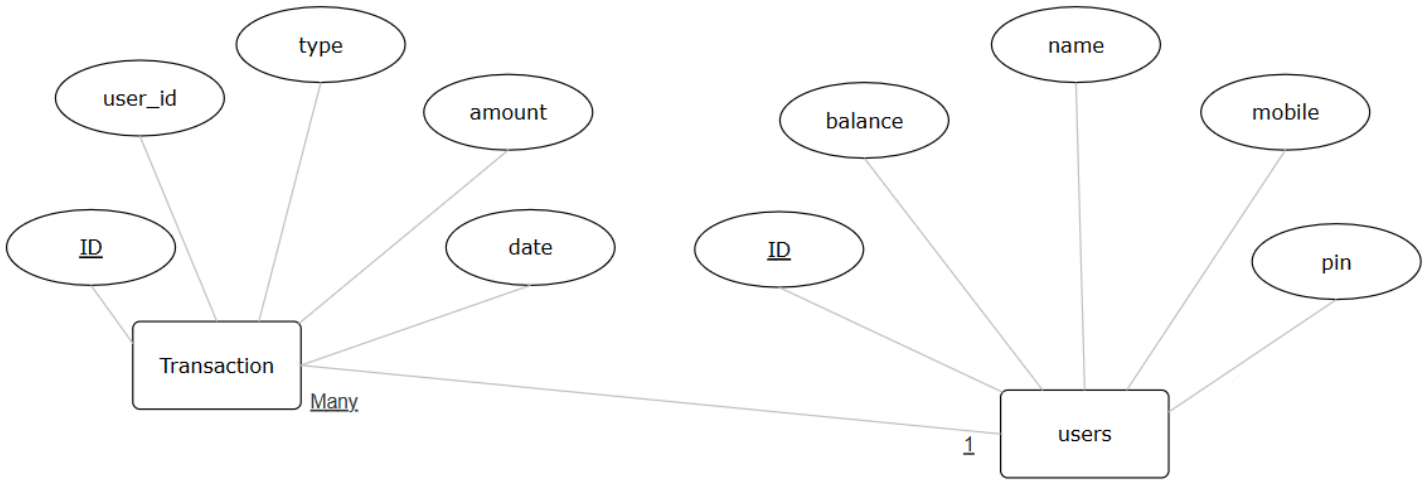

Fig. 2.5. ER Diagram

The ER diagram for the ATM Simulation System represents the logical structure of the database, showing entities, their attributes, and relationships among them. It provides a clear view of how users and transactions interact within the system.

1. Entities and Their Attributes

• **Users**

Attributes: id (PK), name, mobile, pin, balance

Represents bank users who perform transactions using the ATM system.

• **Transaction**

Attributes: id (PK), user_id (FK), type, amount, date

Represents financial operations such as withdrawal, deposit, and balance inquiry performed by users.

2. Relationships

• **Users → Transaction:** One user can perform multiple transactions (1 : M).

• Each transaction is associated with exactly one user through the foreign key (user_id).

3. ER Diagram Summary

- The central entity in the system is the **Users**, which is linked to **Transaction** records.
- The one-to-many relationship ensures that all transactions are properly mapped to their respective users.
- The database structure is simple and normalized, avoiding redundancy and ensuring data consistency.
- This ER design supports efficient transaction management and secure handling of user data in the ATM system.

3. System design

A) Modules

1. Authentication Module

This module handles user login and verification in the ATM system.
Users can:

- Login using Card + PIN
- Login using Mobile Number + PIN
- Authenticate using OTP (cardless withdrawal)
- Use Fingerprint authentication (simulated)
- Access is blocked after multiple incorrect attempts

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2. Transaction Module

This is the core module of the system and handles all banking operations. Users can:

- Withdraw cash
- Deposit money
- Check account balance
- View mini statement
- Change PIN

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3. Account Module

This module manages different types of bank accounts. Users can:

- Select account type (Savings, Current, Salary)
- View account balance
- Perform transactions on selected account
- Maintain account details

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4. UPI QR Module

This module handles modern digital withdrawal using QR code. Users can:

- Generate QR code for withdrawal
- Scan QR using mobile UPI app
- Verify payment
- Complete cardless withdrawal

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5. OTP Withdrawal Module

This module enables cardless withdrawal using OTP. Users can:

- Enter registered mobile number
- Receive OTP
- Verify OTP
- Withdraw cash securely

6. Admin Module
This module is used by the administrator to manage the system. Admin can:

- View all users and transactions
- Add new users
- Reset user PIN
- Block or unblock users
- Monitor system activity

7. System Module (Backend Services)
This module forms the backbone of the ATM Simulation System. It handles internal logic, data flow, and database interaction.
It includes:

- Performing CRUD operations on Users and Transactions
- Managing relationships between Users and Transactions
- Handling database connectivity (JDBC / Hibernate)
- Managing transactions (commit/rollback)
- Ensuring data consistency and security

B) Data Structure of All Modules
We have organized one database named **atm_system** for the system design.
It is implemented using SQLite/MySQL and consists of tables representing users and transactions.

1. Customized Tables
Users Table (users)
Stores information about ATM users.
Attributes:

- id (PK)
- name
- mobile
- pin
- balance

Transaction Table (transaction)
Stores details of all transactions performed by users.
Attributes:

- id (PK)
- user_id (FK)
- type
- amount
- date

Foreign Key

- user_id → References users(id)

2. Relationships Between Tables (ER/Class Diagram)

- **Users → Transaction:** One user can perform multiple transactions (1 : M)
- Each transaction belongs to one user using the foreign key (user_id)

A) Screenshots

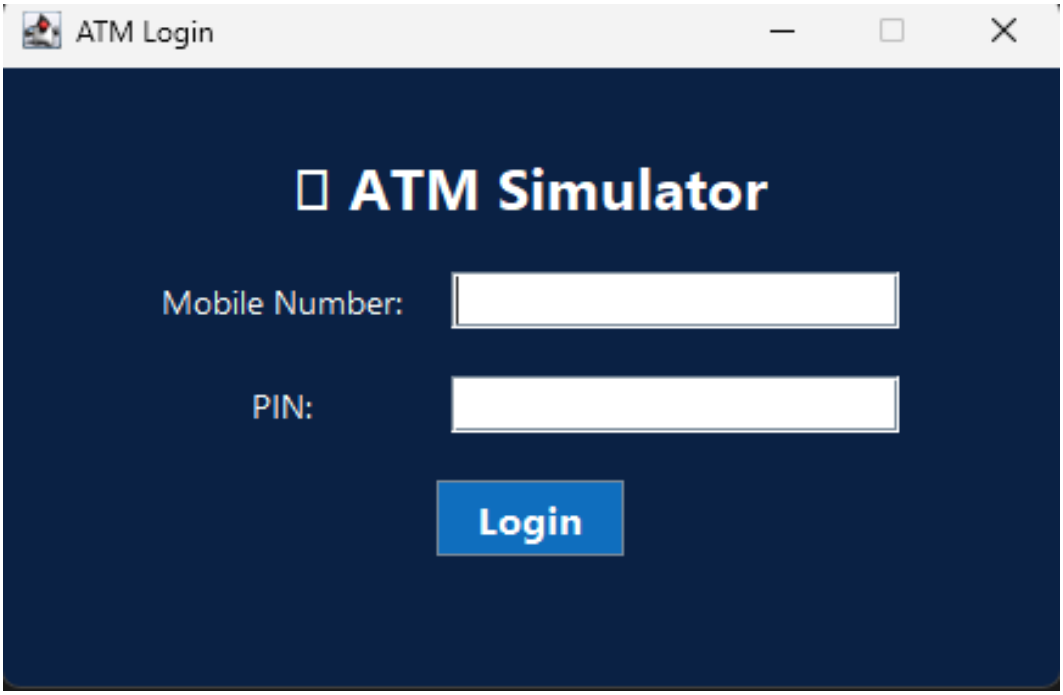

Fig. 3.1. Login Screen

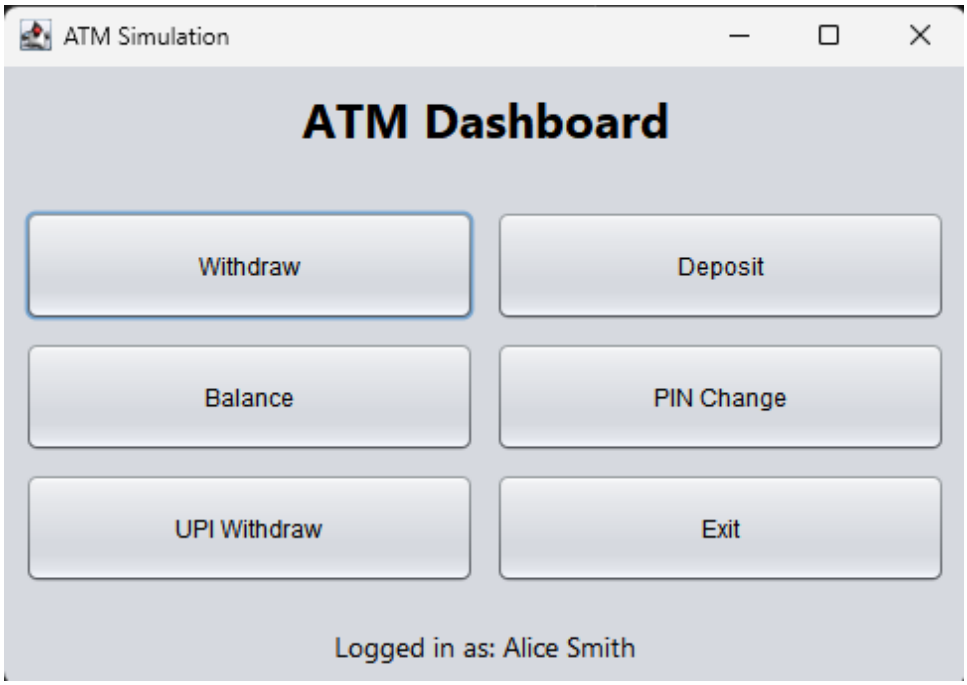

Fig. 3.2. Main screen

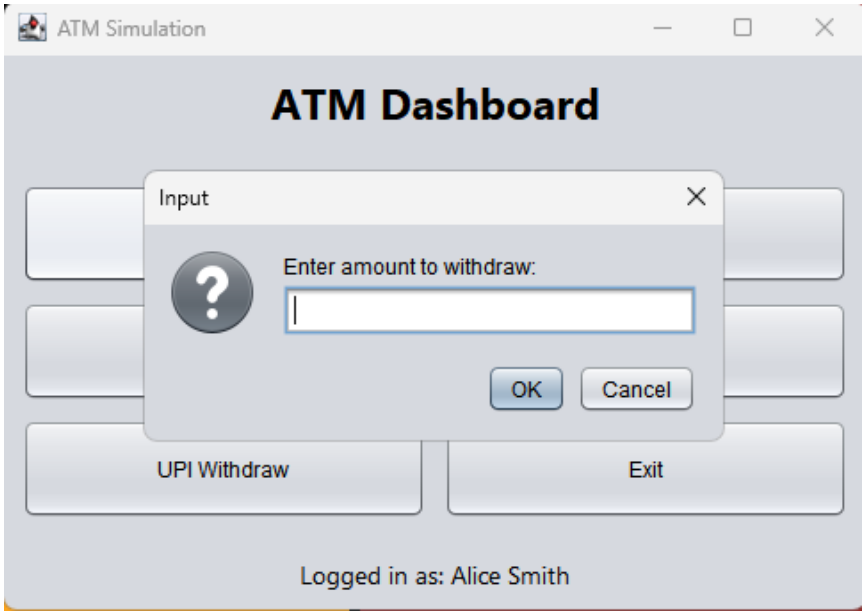

Fig. 3.3. Withdraw

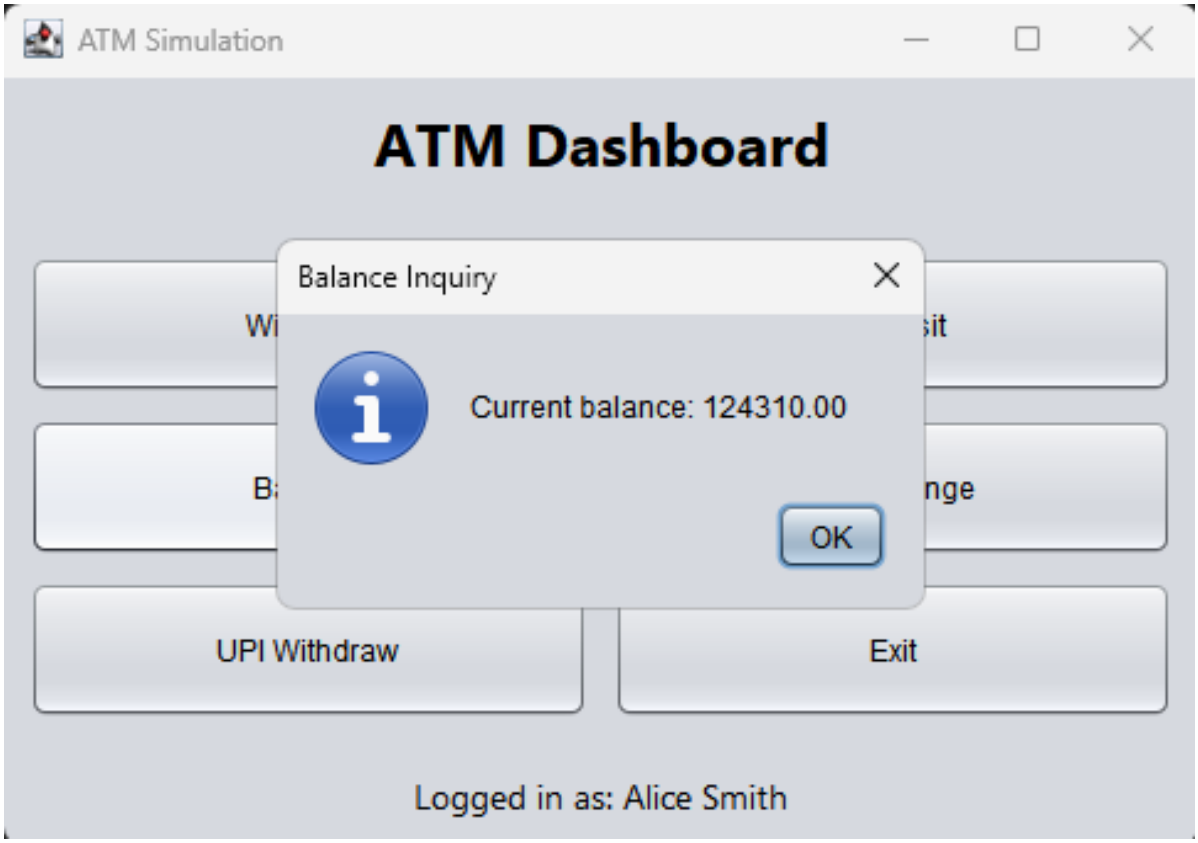

Fig. 3.4. Check Balance

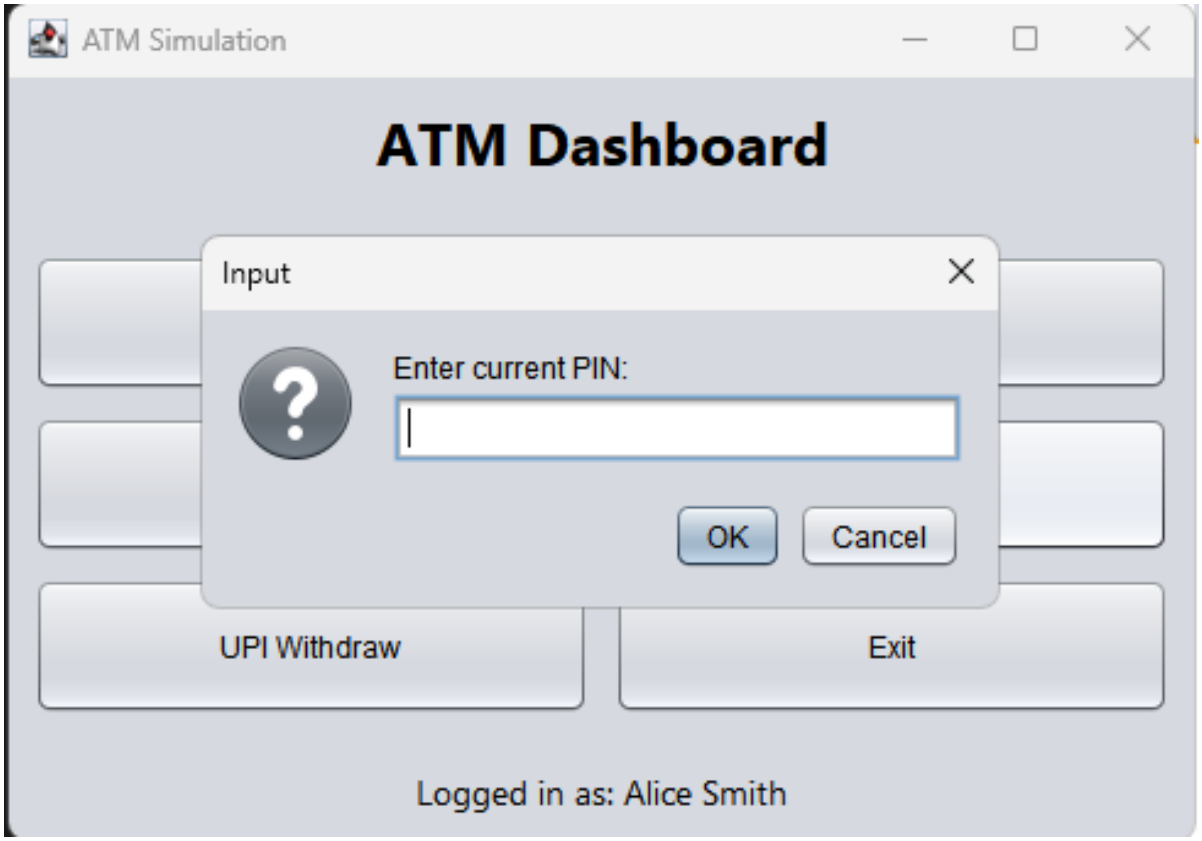

Fig. 3.5. Change PIN

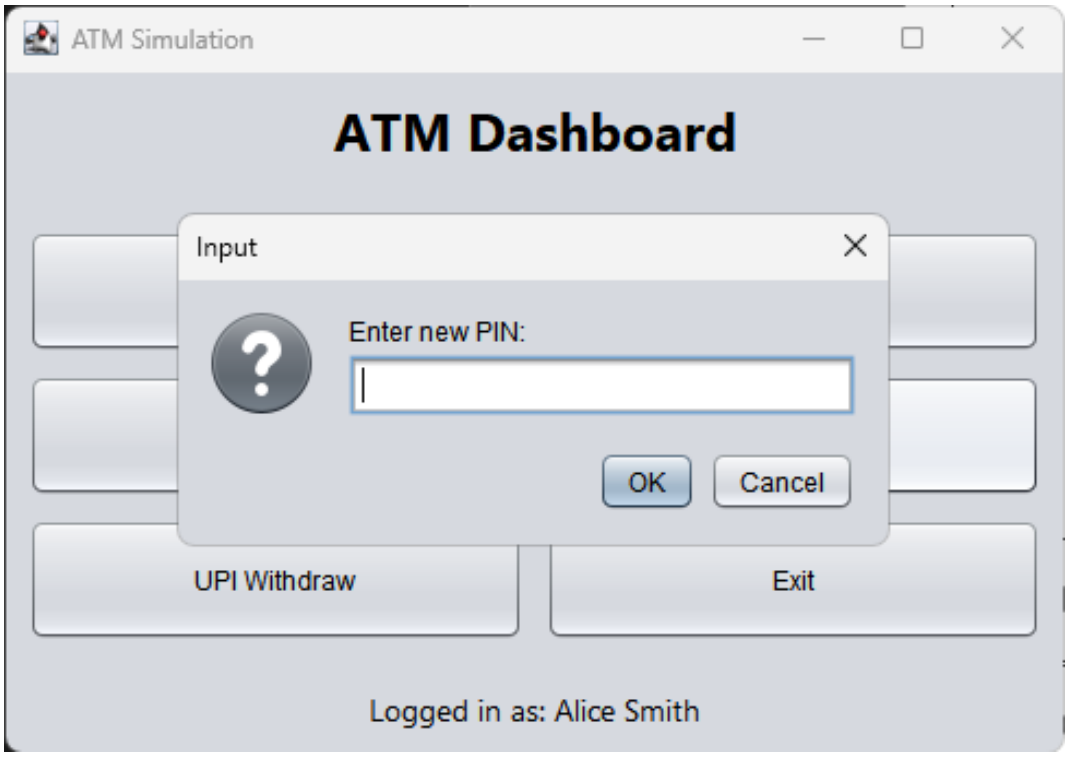

Fig. 3.6 Enter new pin 2 times to change poin

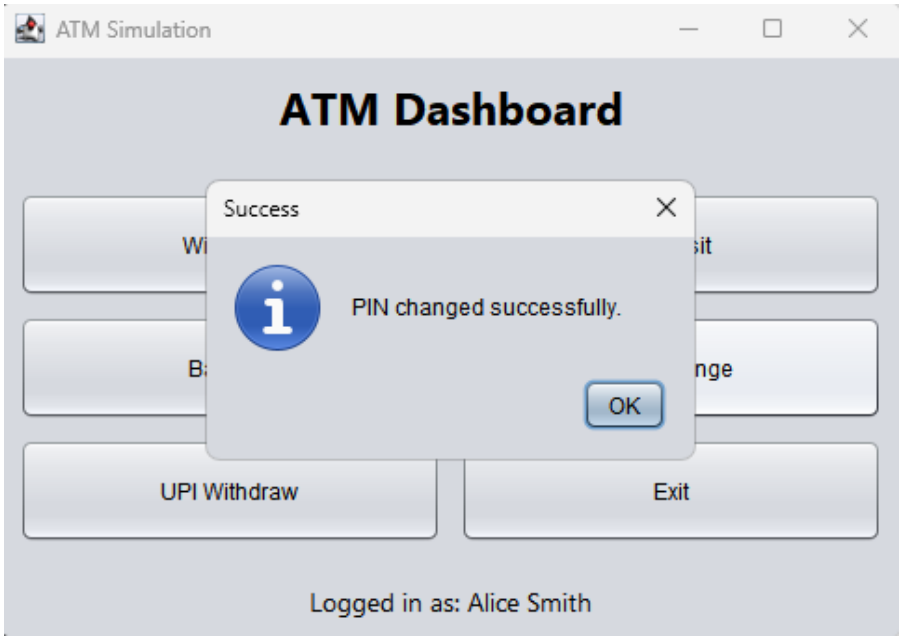

Fig 3.7 : Pin successfully changed

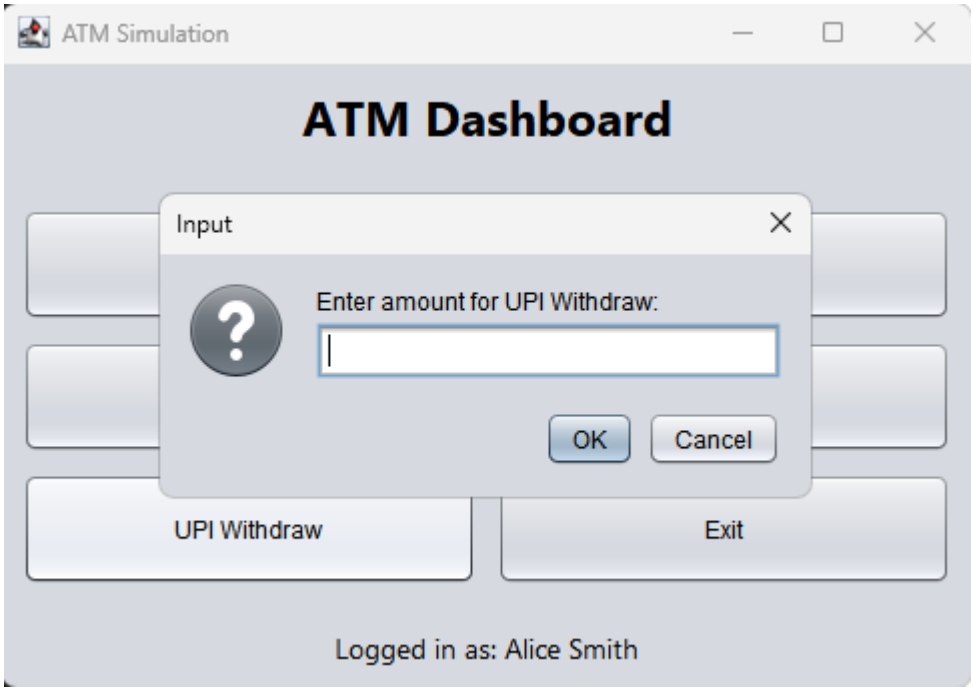

Fig. 3.7. Exiting the session after completing the task

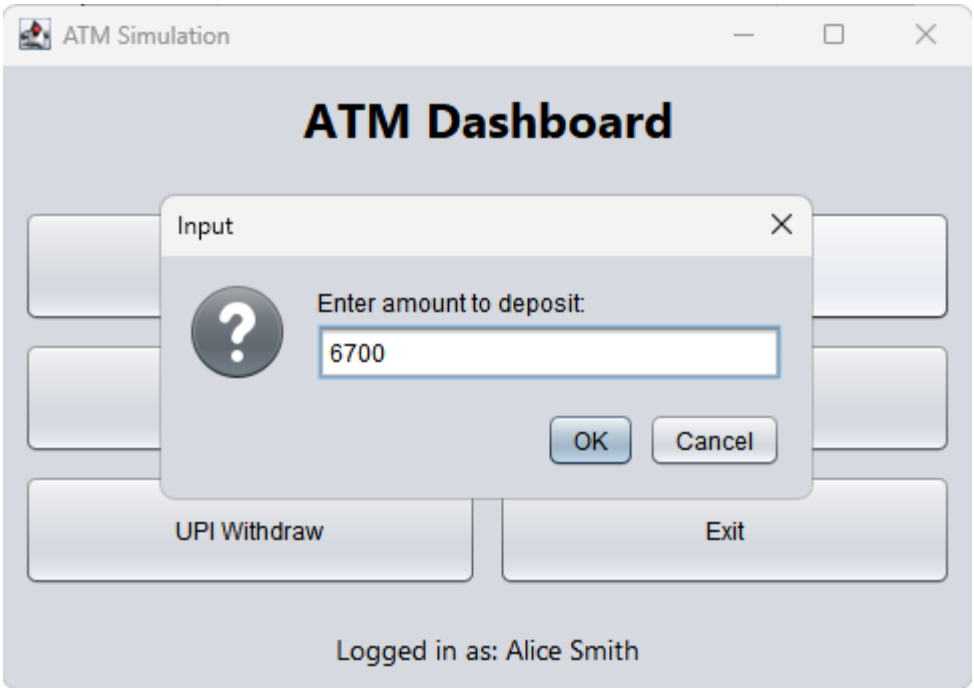

Fig 3.8 Deposit Money

```
mysql> select *from users;
+----+--------------+------------+------+---------+
| id | name         | mobile     | pin  | balance |
+----+--------------+------------+------+---------+
|  1 | Alice Smith  | 9876543210 | 5555 |  124910 |
|  2 | Rahul Sharma | 9123456780 | 2222 |    4916 |
|  3 | Priya Mehta  | 9988776655 | 400  |   56600 |
|  4 | Amit Verma   | 9090909090 | 4444 |    3000 |
+----+--------------+------------+------+---------+
4 rows in set (0.03 sec)

mysql> select *from transaction;
ERROR 1146 (42S02): Table 'atm_db.transaction' doesn't exist
mysql> select *from transactions;
+----+---------+--------------+--------+---------------------+
| id | user_id | type         | amount | date                |
+----+---------+--------------+--------+---------------------+
|  1 |       3 | withdraw     |    500 | 2026-03-21 12:51:58 |
|  2 |       3 | deposit      |  50000 | 2026-03-21 12:52:11 |
|  3 |       3 | upi_withdraw |    400 | 2026-03-21 12:52:31 |
|  4 |       1 | withdraw     |    400 | 2026-03-21 12:56:14 |
|  5 |       1 | upi_withdraw |    700 | 2026-03-21 12:58:46 |
|  6 |       2 | withdraw     |     40 | 2026-03-21 13:04:54 |
|  7 |       2 | upi_withdraw |     44 | 2026-03-21 13:05:42 |
|  8 |       1 | upi_withdraw |    454 | 2026-03-21 13:25:59 |
|  9 |       1 | upi_withdraw |     46 | 2026-03-21 13:27:21 |
| 10 |       1 | withdraw     |    900 | 2026-03-21 13:41:48 |
| 11 |       1 | deposit      |  50000 | 2026-03-21 13:43:26 |
```

Fig 3.9 Database

4. Implementation

The implementation phase involves converting the system design into a functional software application. The ATM Simulation System was developed as a Java-based application using Swing UI (or console-based interface) with SQLite/MySQL for database management. This section describes the technologies, logic, and design used in the implementation.

1. Technology Stack

- Programming Language: Java (JDK 17)
- UI Framework: Java Swing (for graphical interface) / Console-based
- Database: SQLite / MySQL (Relational Database Management System)
- Database Connectivity: JDBC
- IDE: IntelliJ IDEA / Eclipse
- Libraries: ZXing (for QR code generation)

2. Backend Implementation

The backend handles all core logic and database operations using Java and JDBC. CRUD functionalities are implemented for the main entities: Users and Transactions.

Key Functionalities:

- **User Management:**
  Add, view, and manage user details such as name, mobile number, PIN, and balance.

- **Transaction Management:**
  Perform withdrawal, deposit, and balance inquiry operations and store transaction records.

- **Authentication System:**
  Validate user login using PIN, mobile number, OTP, or fingerprint (simulated).

- **UPI QR Withdrawal:**
  Generate QR codes and simulate payment verification for withdrawal.

3. User Interface Implementation

The system uses a menu-driven interface (console or Swing-based ATM UI).

- Users interact through ATM-style screens
- Inputs are captured using Scanner (console) or UI components (Swing)
- Options are displayed similar to real ATM (left-right buttons in UI version)

4. Database Implementation

The database is implemented using SQLite/MySQL with two main tables:

• **Users Table**
Stores user details including id, name, mobile, PIN, and balance.

• **Transaction Table**
Stores transaction details such as id, user_id, type, amount, and date.

Relationships:

• One User → Many Transactions (1 : M)

5. Security Implementation

• **PIN Encryption:** PIN stored securely (can be hashed using SHA-256)
• **Foreign Key Constraint:** Ensures each transaction is linked to a valid user
• **Transaction Management:** Ensures operations are completed fully or rolled back
• **Exception Handling:** Handles invalid inputs and runtime errors
• **Data Validation:** Prevents invalid transactions (e.g., negative withdrawal, insufficient balance)

5. Results and Discussion

The development and implementation of the ATM Simulation System produced several notable outcomes. This section highlights the system's functional effectiveness, performance, and areas for potential improvement based on testing and execution.

1. Results

o Functional Results

- **Authentication Module:**
  Users can successfully log in using PIN, mobile number, OTP, or fingerprint (simulated). Invalid attempts are restricted for security.
- **Transaction Module:**
  Users can perform withdrawal, deposit, balance inquiry, and mini statement operations accurately.
- **Account Module:**
  Multiple account types (Savings, Current, Salary) are handled correctly, and users can select accounts before transactions.
- **UPI QR Module:**
  QR codes are generated successfully, and withdrawal flow works correctly through simulated payment verification.
- **OTP Withdrawal Module:**
  Users can withdraw money without a card using OTP verification, ensuring secure access.
- **Database Operations:**
  All CRUD operations for Users and Transactions execute successfully using JDBC/MySQL, ensuring reliable data storage.

- **Non-Functional Results**

• **Usability:**
The ATM interface (console/Swing) is simple, user-friendly, and easy to navigate.

• **Performance:**
Database queries and transaction processing execute efficiently with quick response time.

• **Reliability:**
Proper exception handling ensures smooth functioning even with invalid inputs.

• **Data Integrity:**
Relationships between Users and Transactions are maintained using foreign key constraints.

• **Scalability:**
The system design allows easy addition of new features such as admin panel, API integration, or mobile banking.

2. Discussion

1. Achievement of Objectives

- All major objectives—such as secure authentication, transaction processing, and cardless withdrawal—were successfully achieved.
- The relationship between Users and Transactions was correctly implemented, ensuring accurate data handling.

2. User Experience

- The system provides a structured and smooth experience through a menu-driven interface.
- ATM-style navigation makes it realistic and easy to use.

3. Challenges Faced

- Designing a realistic ATM UI with left-right button layout was challenging.
- Implementing QR-based UPI withdrawal required proper flow simulation.
- Ensuring secure PIN handling and transaction validation required careful logic implementation.

4. Limitations

- The system uses simulated features (QR payment and fingerprint), not real banking APIs.
- Limited to basic database design (Users and Transactions).
- No real-time network integration with banking systems.

5. Future Enhancements

- Integrate real UPI APIs for live transactions.
- Add biometric hardware integration for fingerprint authentication.
- Develop a web or mobile-based version of the ATM system.
- Implement advanced security features like encryption and fraud detection.
- Add detailed reports and analytics dashboard.

6. Conclusion

The development of the ATM Simulation System has successfully demonstrated the design and implementation of a structured, efficient, and reliable platform for managing banking transactions.

The system enables users to:

- Perform secure transactions such as withdrawal, deposit, and balance inquiry
- Access services using multiple authentication methods (PIN, OTP, and biometric)
- Use modern features like QR-based UPI withdrawal and cardless transactions
- Maintain transaction records efficiently using a database system

Key Achievements Include:

- Successful implementation of core ATM functionalities with accurate transaction handling
- Integration of Users and Transactions ensuring data consistency
- Use of JDBC for reliable database interaction and data persistence
- A modular system design that allows easy scalability and feature expansion
- Proper exception handling and validation ensuring system stability

Overall, the ATM Simulation System provides a realistic and efficient simulation of modern ATM operations. It reduces manual effort, improves user convenience, and ensures secure transaction handling — offering a strong foundation for future enhancements such as real banking integration and advanced security features.

7. Future scope

Although the ATM Simulation System effectively meets its current objectives, several enhancements can be introduced to make the system more advanced, secure, and user-friendly in the future:

1. Web-Based or Advanced UI Interface
Develop a fully functional web or desktop-based interface using JavaFX or Spring Boot.
Provide a modern, responsive, and visually appealing design to improve user experience.

2. Advanced Authentication & Security
Implement enhanced security features such as:

- Multi-factor authentication (OTP + PIN)
- Biometric authentication using real fingerprint devices
- Role-based access control for Admin and Users

3. Real-Time Banking Integration
Integrate real banking APIs and UPI services for live transactions.
Enable real-time account updates and secure payment processing.

4. Notifications & Alerts
Integrate SMS or email notifications for:

- Transaction confirmations
- OTP verification
- Suspicious activity alerts

5. Reporting and Analytics
Generate graphical reports and dashboards showing:

- Transaction history
- User activity
- ATM usage statistics

Provide export options for reports in PDF or Excel format.

6. Advanced Features

- AI-based fraud detection system
- ATM cash management and refill system
- Mobile banking application integration
- Multi-language support for better accessibility

8. Bibliography and References

In preparing this project report on ATM Simulation System, several books, research papers, and online resources were referred to. These materials provided valuable insights into system design, database management, software development methodologies, and banking system concepts.

Books

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